Quadriceps and hamstrings strength reference values for athletes with and without ACL reconstruction who play popular pivoting sports, including soccer, basketball and handball: a scoping review.

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Abstract

Objective: To synthesise and present reference values for quadriceps and hamstring strength tests in healthy athletes who play pivoting sports and athletes with anterior cruciate ligament reconstruction (ACLR) who play pivoting sports.

Design: scoping review.

Literature search: We searched PubMed, the Cochrane Library, MEDLINE, Embase and Web of Science until January 26, 2021.

Study selection criteria: We included reference values in two different categories: I) quadriceps and hamstring strength test outcomes from healthy pivoting sport athletes and II) quadriceps and hamstring strength test outcomes from pivoting sport athletes with ACLR at a specific time point during rehabilitation.

Data synthesis: We performed a qualitative synthesis for reference values from isokinetic (at 60, 180 and 300 °/sec) and isometric quadriceps and hamstring strength tests. We summarised the data for type of sport, sex, sport participation level and age group.

Results: Of the 42 included studies, 26 reported reference values from healthy soccer players, 4 from healthy basketball players, 4 from healthy handball players and 11 from other healthy pivoting sport athletes. LSI dominant/non-dominant (LSI-D/ND) ranged from 98% to 114% for healthy athletes. Six studies reported reference values from pivoting sport athletes with ACLR at a specific time point during rehabilitation. After 7 months and beyond, strength values for athletes with ACLR are comparable to those of healthy pivoting sport athletes.

Conclusion: This scoping review summarises quadriceps and hamstrings strength reference values for athletes who play the most common pivoting sports, including soccer, basketball and handball.

Key words: anterior cruciate ligament reconstruction, reference values, muscle strength

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Introduction

Quadriceps and hamstring strength tests are important when evaluating rehabilitation progression and making return to sport (RTS) decisions after anterior cruciate ligament reconstruction (ACLR).[6,35,48] Athletes who play pivoting sports and who pass strength tests as part of a RTS testing protocol are at lower risk for a second ACL injury, underscoring the importance of these tests.[19,20,32]

The Limb Symmetry Index (LSI) – a comparison between the operated and non-operated limb - is often used to guide RTS decisions. However, clinicians and researchers have long raised concerns about using the non-operated limb as a reference, as the non-operated limb also detrains and loses strength after injury and surgery.[57] As a result, LSI might overestimate operated knee function and not be sensitive enough to alert clinicians and athletes to a high risk of second ACL injury.[57]

Using both LSI and pre-injury strength values to guide rehabilitation progression could be a solution, but pre-injury data are often unavailable in everyday practice. Strength test reference values from healthy athletes who play pivoting sports could solve this problem, but a comprehensive overview of reference values for different sports is currently lacking.

We aimed to synthesise and present reference values for quadriceps and hamstring strength tests from athletes with and without ACLR who play pivoting sports. The goal of our scoping review was to present information to help clinicians understand what 'normal' quadriceps and hamstrings strength is for the athletes they work with: what to expect from healthy players without ACLR, and what to expect during rehabilitation and RTS progressions after ACLR. We present reference values for different pivoting sports, sexes, sport participation levels (elite and non-elite) and age groups.

Methods

This scoping review was conducted and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-analysis Extension for Scoping Reviews (PRISMA-ScR).[54]

Eligibility criteria

Inclusion and exclusion criteria were specified for two different categories of reference values: I) quadriceps and hamstring strength test outcomes from pivoting sport athletes with ACLR at a specific time point during rehabilitation, from three months until RTS, and II)

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quadriceps and hamstring strength test outcomes from healthy pivoting sport athletes, which can be used as RTS criteria (**TABLE 1**).

Insert TABLE 1.

Search strategy

On the 26th of January 2021 a systematic literature search was performed by an academic librarian. PubMed, the Cochrane Library, MEDLINE, Embase and Web of Science were searched from database inception to identify relevant articles using keywords specified for the database (Supplementary Material 1). In addition, a hand search of the reference lists of meta-analyses and (systematic) reviews was conducted to identify additional studies not found in the primary search.

All database records were exported to the Rayyan QCRI webapp, in separate files for quadriceps and hamstring strength (www.rayyan.qcri.org).[45] Duplicates were removed from each file.

Study selection

Two authors (NvM, WvdW) independently screened titles and abstracts for eligibility. For all potentially eligible studies, a full-text version was reviewed by two authors (WvdW, NvdH). Any disagreements in both steps were resolved by consensus. After this, both quadriceps and hamstring strength Rayyan QCRI files were combined, and duplicates were removed.

Data synthesis

One author (NvM) extracted all relevant study characteristics and reference values data. Study characteristics included author and year of publication, population characteristics, test details and test outcome variables. Reference values were categorized as pivoting sport athlete with or without ACLR, type of sport, sex, sport level (elite or non-elite), and age group (adolescent 16-19 years old or young adult 20-35 years old).[60]

Test outcome variables reported as reference values were peak torque (PT), peak torque normalized to body weight (PT/BW), total work (TW; only reported for an endurance test with at least ten repetitions and when test range was described), LSI calculated as ((value operated leg)/(value non-operated leg))*100% for athletes with ACLR, LSI-D/ND calculated as ((value dominant leg)/(value non-dominant leg))*100% for healthy athletes,[37] concentric hamstring to quadriceps ratio (HQR), dynamic control (eccentric hamstring to concentric quadriceps) ratio (DCR) and maximum isometric voluntary contraction (MIVC). LSIs and

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ratios were extracted or calculated from the individual studies. When an LSI based on both PT and PT/BW were available, only the LSI based on PT/BW was reported.

When three or more studies reported on the same sport, we created a reference value table for the sport. We grouped data from all other sports (reported in two or fewer studies) in one extra reference value table.

All outcome variables reported in two or more studies were presented as a weighted mean with weighted standard deviation. The weighted mean was calculated as: sum of (study mean * study sample size) divided by the sum of all study sample sizes.

The weighted standard deviation was calculated as: sum of (study variance * study sample size) divided by sum of all study sample sizes. Study variance was calculated as: |study mean minus weighted mean|.

Outcome variables reported in a single study were displayed as mean with standard deviation. Standard deviation was not available if we calculated LSI or ratio from data extracted from a specific study.

Results

Study selection

After removing duplicates, the systematic literature search yielded 712 articles for quadriceps strength and 246 articles for hamstring strength. After screening title/abstract and full-text for eligibility, 924 articles were excluded, leaving 32 articles for quadriceps and hamstring strength combined. Hand searching reference lists of meta-analysis and (systematic) reviews provided 10 additional articles; 42 articles were included for data synthesis (**FIGURE 1**).

Insert FIGURE 1.

Overview of strength testing protocols

Data on study characteristics are described in Supplementary Material 2. All included studies performed strength measurements using an isokinetic dynamometer, with a Biodex or Cybex used most often (55% and 33% respectively). Thirty-nine studies measured isokinetic strength only, 2 measured isometric strength only, and 1 measured isokinetic and isometric strength.

Isokinetic strength testing protocols

Two studies measured concentric quadriceps strength only, 28 studies measured concentric quadriceps and hamstring strength, 6 studies measured concentric quadriceps and

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concentric plus eccentric hamstring strength, and 4 studies measured concentric plus eccentric quadriceps and hamstring strength.

Thirty-six studies performed concentric strength measurements at 60 degrees per second, usually with 3 or 5 repetitions (39% and 42% respectively). Seventeen studies reported strength measurements at 180 and/or 300 degrees per second, and typically 3 and 5 repetitions (29% and 35% respectively). One study, which reported TW as an outcome parameter, performed more than 10 repetitions at these higher speeds. [63] Eccentric strength was typically measured over 5 repetitions (70%). Range of motion was not specified in 33% of all isokinetic studies. In those that reported range of motion 100-0° of flexion and 90-0° of flexion were mentioned in 15% and 33% respectively. Rest between sets and whether gravity correction was performed or not, was usually not described.

Isometric strength testing protocols

Isometric strength measurements were all performed at a different knee angle (45, 70 or 90° of flexion) and with different duration (2, 3 or 4 seconds).

Quadriceps and hamstrings strength test results: soccer

Strength results from healthy soccer players were reported in 26 studies.[5,8-11,13-16,18,21,22,26,31,38,39,44,46,49-51,53,55,56,59,62] There were data from 1987 soccer players (84% males) aged a mean 24.0 ± 2.1 years (**TABLE 2)**.

Quadriceps and hamstrings strength test results: basketball

Strength results from healthy basketball players were reported in 4 studies.[17,31,38,52] There were data from 99 healthy basketball players (74% males) aged a mean 22.5 ± 0.99 years (**TABLE 3).**

Quadriceps and hamstrings strength test results: handball

Strength results from healthy handball players were reported in 4 studies.[14,17,33,49] There were data from 310 handball players (5% males) aged a mean 21.3 ± 1.3 years (**TABLE 4**).

Quadriceps and hamstrings strength test results: other pivoting sports

All other pivoting sports (e.g. volleyball, hockey, futsal, American football, Australian football, judo, alpine skiing, and modern ballet) were grouped with unspecified pivoting sports in **TABLE 5**. Strength results from other pivoting sports were reported in 11 studies. [1,3,9,12,14,28,41-43,47,63] There were data from 1566 pivoting sport athletes (92% males) aged a mean 21.7 ± 1.9 years.

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Quadriceps and hamstrings strength test results: pivoting sport athletes with ACLR

Strength results from six studies of pivoting sport athletes with ACLR at a specific time point during rehabilitation were combined in **TABLE 6**.[7,27,30,42,56,61] Strength tests were completed at three, four, six, seven, nine and ten months after ACLR for 816 pivoting sport athletes with ACLR (80% males) aged a mean 21.8 ± 3.0 years. After 7 months and beyond, strength values for athletes with ACLR were comparable to those of healthy pivoting sport athletes.

Insert TABLES 2, 3, 4, 5, and 6.

Discussion

We aimed to synthesise and present reference values for quadriceps and hamstring strength tests for pivoting sport athletes with and without ACL reconstruction. We presented separate results for types of pivoting sports (including soccer, basketball and handball), sex, sport participation levels and age groups. We aim for the reference values to help guide clinicians regarding what is 'normal' strength for pivoting sport athletes, and what to expect during rehabilitation and RTS progressions after ACLR.

Strength testing modes

Isokinetic dynamometry is the gold standard for strength tests. We chose to report reference values at 60, 180 and 300°/sec. This was based on a recent Delphi study, in which experts (physical therapists, orthopedic surgeons and scientists) recommended a protocol with concentric knee extension and flexion testing, including five repetitions at 60°/sec, 20 repetitions at 180°/sec and 15 repetitions at 300°/sec, with 60 seconds rest between sets.[25] Isokinetic dynamometry is unfortunately only accessible for a small proportion of clinicians due to its non-portability and high cost.[36]

When isokinetic dynamometry is not available, (belt-stabilized) hand-held dynamometry could be a reliable alternative for measuring quadriceps and hamstring strength.[24,29] Hand-held dynamometry strength measurements could be compared to maximum voluntary isometric contraction (MIVC) reference values in our scoping review. However, the MIVC reference values in our review are all measured with isokinetic dynamometry, and the validity of hand-held dynamometry measurements, when judged against isokinetic dynamometry measurements, remains debated.[24,40]

Strength tests during rehabilitation after ACLR

Because RTS is a continuous process that starts from the beginning of rehabilitation, it is important to evaluate strength multiple times throughout rehabilitation.[4,35] Regular testing

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allows for easy modification of strength training programs, tailored to the athlete's strength training status and needs. The reference values from our scoping review may help guide goal setting and evaluation during rehabilitation, and guide clinicians to ideal and realistic expectations. Although reference values during rehabilitation are reported for three, four, six, seven, nine and ten months postoperatively, consider approximately 2 months is needed between measurements to observe clinically-meaningful changes.[4] There are of course individual differences in how fast athletes make meaningful progress, but it is important not to test too little or too often, to keep athletes motivated. With adequate rehabilitation, it is possible to strive for strength values comparable to those of healthy pivoting sport athletes from 7 months after ACLR.[56]

Strength tests as part of the RTS decision

Return to sport decisions after ACLR should be based on a battery of tests.[2] However, the specific content of such a battery is the subject of ongoing and strident debate. Conflicting research in the ACL field compounds a complex debate. One key issue is that some studies include athletes who do not return to pivoting sports—a main confounder when interpreting the results, because not returning to pivoting sports virtually eliminates the risk for a second ACL injury.[58] Therefore, when reading studies about the association between RTS tests and second ACL injuries one should be aware of this. When eliminating the pivoting sport participation confounding factor, quadriceps strength test results are associated with ACL and knee re-injuries, and cannot be denied as an important part of an RTS test battery.[2,19,20,32]

Clinical relevance

If pre-injury values are unavailable for injured athletes, or previous season values are unavailable for healthy athletes, clinicians could choose to use the reference tables from our scoping review as a guide to the expected quadriceps and hamstring strength values for an individual athlete specifically. Both PT - MIVC and PT/BW - MIVC/BW are reported as absolute reference values. We suggest using PT/BW or MIVC/BW for comparing between athletes, because they better account for the athlete's body weight. When these metrics are unavailable, PT or MIVC are the best alternative.

Besides using absolute reference values, it is important to consider using an LSI. However, because LSI-D/ND is used for reporting differences between the dominant leg and the non-dominant leg in healthy pivoting sport athletes, we advise clinicians to use the same measure for supporting pivoting sport athletes with ACLR when they are making RTS decisions. The LSI-D/ND ranged from 98% to 114% for healthy athletes. We suggest clinicians use this as a benchmark for pivoting sport athletes with ACLR, instead of an LSI>90% as often advised.[34]

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When athletes with ACLR do not meet the expected absolute values and have an LSI-D/ND below the healthy reference, consider additional strength training before return to pivoting sport.

Limitations

We provided an extensive and detailed overview of quadriceps and hamstring strength absolute and LSI reference values. However, there will always be outliers in each set of athletes, as reference values are based on a Gaussian curve.

We used broad selection criteria to maximise generalisability. However, due to sparse data in the group of athletes with ACLR, we were not able to present separate results for different graft types. Graft type could affect strength at different stages of rehabilitation. Therefore, we suggest interpreting reference values for pivoting sport athletes with ACLR as minimum requirements and suggest striving for higher values earlier in the rehabilitation process. It is probably better to use these reference values together with reference values of healthy pivoting sport athletes, also because we do not have insight into postoperative rehabilitation protocols.

Although we present these strength test reference values to help clinicians judge what 'normal' strength is for pivoting sport athletes, is has not yet been investigated whether there is an association between meeting reference values and second ACL injuries.

Conclusion

We synthesise and present absolute quadriceps and hamstrings strength reference values for pivoting sport athletes with and without ACLR. Data from 42 articles are organised by type of sport (including soccer, basketball, handball), sex, sport participation level and age group. Besides using absolute reference values, the LSI-D/ND is valuable to use for RTS decisions. This LSI-D/ND ranged from 98% to 114% for healthy athletes.

Key points

Findings: Detailed reference values are presented for athletes with and without ACLR who play the most common pivoting sports (including soccer, basketball and handball).

Implications: Quadriceps and hamstring strength test reference values help clinicians judge what 'normal' strength for healthy pivoting sport athletes is, and what to expect during rehabilitation and RTS progressions after ACLR.

Caution: The association between meeting reference values and (safe) RTS after ACLR requires more study.

STUDY DETAILS

Data sharing statement

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All data relevant to the study are included in the article or are available as supplementary files.

Please ensure that no patient-identifiable data are available.

Patient involvement statement

There is no patient or public involvement in this manuscript.

Author contributions

- Design: NvM, WvdW, NvdH
- Study selection: NvM, WvdW, NvdH
- Data synthesis: NvM
- Drafting article: NvM
- Revising article: NvM, WvdW, NvdH

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TABLE 1. Inclusion and exclusion criteria for reference values of healthy and ACL-reconstructed pivoting sport athletes.

Inclusion	Exclusion
Population	
Adolescents (mean age 16-19 years old) or young adults (mean age 20-35 years old)	
Male or female athletes	
Elite or recreational athletes	Healthy: History of ACL injury or surgery in the past, or other lower extremity or lower back injury when tested ACLR: ACL revision surgery or contralateral ACL injury in the past
Athletes performing pivoting sports (clear sports description, Tegner ≥6 or level I/II sports*)	
ACLR: Potential concomitant MCL or LCL injuries	ACLR: Concomitant PCL injuries
ACLR: Potential concomitant meniscal or cartilage injuries	
Outcome	
Strength tests for quadriceps or hamstrings with isokinetic dynamometer (isometric or concentric/eccentric at 60-180-300°/sec[25]) or hand-held dynamometer	
	No separate results for male and female athletes
Absolute values or limb symmetry indices as outcome	No separate results for elite or recreational athletes
ACLR: Tests performed at a specific time point during rehabilitation from 4 months until the moment of return to sport	ACLR: Tests performed more than a year after ACLR
Publication type	
All original research types	Meta-analyses, systematic or narrative reviews, congress abstracts, posters
Language: English, Dutch, or German	

*level I sports: sports with frequent pivoting movements (e.g. football, handball, basketball); level II sports: sports with lateral movements, and less pivoting than level I sports (e.g. racket sports, alpine skiing).[23]

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TABLE 2. Quadriceps and hamstring strength reference values from healthy soccer players.[5,8-11,13-16,18,21,22,26,31,38,39,44,46,49-51,53,55,56,59,62]

Sex	Sport level	N, age	Isokinetic 60°/sec	Isokinetic 180°/sec	Isokinetic 300°/sec	Isometric 90°
Male	Elite	58 adolescents, 17.1 ± 0.8 years old.[16,22,46]	Quadriceps PT: 182 ± 28 Nm LSI-D/ND: 102 ± 15 % Hamstrings PT: 97 ± 18 Nm LSI-D/ND: 108 ± 18 % PT eccentric: 151 ± 29 Nm Ratios DCR: 84 %*	Quadriceps PT: 145 ± 13 Nm LSI-D/ND: 99 %* PT eccentric: 244 ± 42 Nm Hamstrings PT: 87 ± 15 Nm LSI-D/ND: 98 %* PT eccentric: 138 ± 21 Nm Ratios HQR: 61 ± 9 % DCR: 96 ± 20 %	Quadriceps PT: 122 ± 14 Nm LSI-D/ND: 99 %* PT eccentric: 249 ± 41 Nm Hamstrings PT: 72 ± 9 Nm LSI-D/ND: 99 %* PT eccentric: 141 ± 23 Nm Ratios HQR: 61 ± 7 % DCR: 117 ± 23 %	Quadriceps MIVC: 409 ± 78 N Hamstrings MIVC: 173 ± 38 N
		1499 young adults, 24.9 ± 1.2 years old.[5,8,10,11,13-15,18,21,22,26,38,39,50,51,55,59,62]	Quadriceps PT: 239 ± 16 Nm PT/BW: 317 ± 10 Nm/kg LSI-D/ND: 99 ± 2 % PT eccentric: 299 ± 12 Nm Hamstrings PT: 138 ± 4 Nm PT/BW: 178 ± 10 Nm/kg LSI-D/ND: 102 ± 3 % PT eccentric: 187 ± 19 Nm PT/BW eccentric: 260 ± 26 Nm/kg Ratios HQR: 60 ± 3 % DCR: 75 ± 3 %	Quadriceps PT: 168 ± 14 Nm PT/BW: 256 ± 10 Nm/kg LSI-D/ND: 103 ± 0 % PT eccentric: 249 ± 40 Nm Hamstrings PT: 106 ± 7 Nm PT/BW: 162 ± 5 Nm/kg LSI-D/ND: 102 ± 0 % PT eccentric: 154 ± 1 Nm Ratios HQR: 62 ± 6 % DCR: 101 ± 3 %	Quadriceps PT: 134 ± 6 Nm PT/BW: 187 ± 1 Nm/kg LSI-D/ND: 102 ± 0 % PT eccentric: 256 ± 15 Nm Hamstrings PT: 93 ± 7 Nm PT/BW: 133 ± 1 Nm/kg LSI-D/ND: 101 ± 4 % PT eccentric: 162 ± 4 Nm Ratios HQR: 68 ± 6 % DCR: 126 ± 5 %	
	Non-elite	10 adolescents, 17.8 ± 0.1 years old.[31]		Quadriceps PT: 125 ± 15 Nm PT/BW: 205 ± 20 Nm/kg Hamstrings PT: 105 ± 7 Nm PT/BW: 170 ± 10 Nm/kg Ratios HQR: 86 ± 7 %	Quadriceps PT: 90 ± 15 Nm PT/BW: 150 ± 20 Nm/kg Hamstrings PT: 70 ± 5 Nm PT/BW: 115 ± 5 Nm/kg Ratios HQR: 82 ± 10 %	

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		106 young adults, 23.5 ± 1.5 years old.[9,10,53,56]	Quadriceps PT: 225 ± 0 Nm PT/BW: 276 ± 41 Nm/kg LSI-D/ND: 106 ± 1 % PT eccentric: 310 ± 10 Nm PT/BW eccentric: 269 ± 45 Nm/kg Hamstrings PT: 130 ± 5 Nm PT/BW: 157 ± 23 Nm/kg LSI-D/ND: 106 ± 6 % PT eccentric: 150 ± 10 Nm PT/BW eccentric: 156 ± 35 Nm/kg Ratios HQR: 61 ± 3 % DCR: 71 ± 7 %	Quadriceps PT: 150 ± 3 Nm Hamstrings PT: 105 ± 7 Nm Ratios HQR: 71 ± 8 %	Quadriceps PT: 125 ± 5 Nm PT/BW: 127 ± 23 Nm/kg LSI-D/ND: 106 %* Hamstrings PT: 90 ± 7 Nm PT/BW: 98 ± 19 Nm/kg LSI-D/ND: 114 %* Ratios HQR:79 ± 1 %	
Female	Elite	213 young adults, 21.3 ± 0.6 years old.[14,49]	Quadriceps PT: 149 ± 3 Nm PT/BW: 232 ± 36 Nm/kg LSI-D/ND: 101 %* Hamstrings PT: 87 ± 1 Nm PT/BW: 136 ± 21 Nm/kg LSI-D/ND: 103 %* Ratios HQR: 59 ± 1 %		Quadriceps PT: 83 ± 12 Nm Hamstrings PT: 60 ± 9 Nm Ratios HQR: 72 ± 11 %	
	Non-elite	101 young adults, 20.3 ± 4.1 years old.[44]	Quadriceps PT: 88 ± 15 Nm LSI-D/ND: 101 %*	Quadriceps PT: 60 ± 10 Nm LSI-D/ND: 101 %*		

DCR= dynamic control ratio=eccentric hamstring strength to concentric quadriceps strength ratio; HQR= concentric hamstring strength to concentric quadriceps strength ratio; LSI-D/ND=limb symmetry index calculated as ((value dominant leg)/(value non-dominant leg))*100%; MIVC=maximum isometric voluntary contraction; PT=peak torque; PT/BW=peak torque normalized to body weight.
*Standard deviation not available.

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TABLE 3. Quadriceps and hamstring strength reference values from healthy basketball players.[17,31,38,52]

Sex	Sport level	N, age	Isokinetic 60°/sec	Isokinetic 180°/sec	Isokinetic 300°/sec
Male	Elite	73 young adults, 22.7 ± 0.6 years old.[38,52]	Quadriceps PT: 289 ± 3 Nm PT/BW: 321 ± 47 Nm/kg LSI-D/ND: 105 %* Hamstrings PT: 157 ± 8 Nm PT/BW: 206 ± 35 Nm/kg LSI-D/ND: 107 %* Ratios HQR: 55 ± 3 %	Quadriceps PT: 190 ± 12 Nm PT/BW: 173 ± 31 Nm/kg LSI-D/ND: 98 %* Hamstrings PT: 107 ± 7 Nm PT/BW: 145 ± 27 Nm/kg LSI-D/ND: 100 %* Ratios HQR: 58 ± 9 %	Quadriceps PT: 147 ± 27 Nm Hamstrings PT: 82 ± 19 Nm Ratios HQR: 56 ± 10 %
Female	Elite	14 young adults, 24.4 ± 2.6 years old.[17]	Quadriceps PT: 185 ± 15 Nm PT/BW: 250 ± 15 Nm/kg Hamstrings PT: 100 ± 10 Nm PT/BW: 150 ± 8 Nm/kg Ratios HQR: 57 ± 9 %	Quadriceps PT: 120 ± 10 Nm PT/BW: 170 ± 8 Nm/kg Hamstrings PT: 55 ± 5 Nm PT/BW: 90 ± 6 Nm/kg Ratios HQR: 55 ± 10 %	Quadriceps PT: 75 ± 10 Nm PT/BW: 120 ± 8 Nm/kg Hamstrings PT: 30 ± 4 Nm PT/BW: 50 ± 8 Nm/kg Ratios HQR: 51 ± 10 %
	Non-elite	12 young adults, 20.1 ± 0.4 years old.[31]		Quadriceps PT: 105 ± 8 Nm PT/BW: 190 ± 20 Nm/kg Hamstrings PT: 75 ± 7 Nm PT/BW: 140 ± 10 Nm/kg Ratios HQR: 69 ± 15 %	Quadriceps PT: 60 ± 5 Nm PT/BW: 105 ± 10 Nm/kg Hamstrings PT: 47 ± 7 Nm PT/BW: 85 ± 8 Nm/kg Ratios HQR: 78 ± 8 %

HQR= concentric hamstring strength to concentric quadriceps strength ratio; LSI-D/ND=limb symmetry index calculated as ((value dominant leg)/(value non-dominant leg))*100%; PT=peak torque; PT/BW=peak torque normalized to body weight (Nm/kg).
*Standard deviation not available.

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TABLE 4. Quadriceps and hamstring strength reference values from healthy handball players.[14,17,33,49]

Sex	Sport level	N, age	Isokinetic 60°/sec	Isokinetic 180°/sec	Isokinetic 300°/sec
Male	Elite	17 young adults, 25.9 ± 4.1 years old.[14]	Quadriceps PT: 266 ± 51 Nm Hamstrings PT: 163 ± 18 Nm Ratios HQR: 63 ± 12 %		Quadriceps PT: 181 ± 36 Nm Hamstrings PT: 113 ± 22 Nm Ratios HQR: 63 ± 9 %
Female	Elite	293 young adults, 21.1 ± 0.9 years old.[14,17,33,49]	Quadriceps PT: 169 ± 5 Nm PT/BW: 244 ± 5 Nm/kg LSI-D/ND: 100 %* Hamstrings PT: 95 ± 2 Nm PT/BW: 138 ± 2 Nm/kg LSI-D/ND: 103 %* Ratios HQR: 57 ± 1 %	Quadriceps PT: 110 ± 7 Nm PT/BW: 170 ± 8 Nm/kg Hamstrings PT: 40 ± 7 Nm PT/BW: 90 ± 6 Nm/kg Ratios HQR: 55 ± 10 %	Quadriceps PT: 91 ± 16 Nm PT/BW: 110 ± 8 Nm/kg Hamstrings PT: 53 ± 18 Nm PT/BW: 50 ± 6 Nm/kg Ratios HQR: 61 ± 5 %

HQR= concentric hamstring strength to concentric quadriceps strength ratio; LSI-D/ND=limb symmetry index calculated as ((value dominant leg)/(value non-dominant leg))*100%; PT=peak torque; PT/BW=peak torque normalized to body weight.
*Standard deviation not available.

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TABLE 5. Quadriceps and hamstring strength reference values from other healthy pivoting sport athletes (volleyball, hockey, futsal, American football, Australian football, judo, alpine skiing, modern ballet).[1,3,9,12,14,28,41-43,47,63]

Sex	Sport level	N, age	Isokinetic 60°/sec	Isokinetic 180°/sec	Isokinetic 300°/sec	Isometric 70°
Male	Elite	20 adolescents, 17.0 ± 0.5 years old.[3]	Quadriceps PT/BW: 342 ± 40 Nm/kg LSI-D/ND: 99 %* Hamstrings PT/BW: 163 ± 16 Nm/kg LSI-D/ND: 103 %* Ratios HQR: 48 ± 6 %		Quadriceps PT/BW: 192 ± 20 Nm/kg LSI-D/ND: 102 %* Hamstrings PT/BW: 127 ± 17 Nm/kg LSI-D/ND: 102 %* Ratios HQR: 67 ± 12 %	
		1361 young adults, 22.5 ± 1.2 years old.[3,14,28,41,43,63]	Quadriceps PT: 309 ± 3 Nm PT/BW: 289 ± 3 Nm/kg LSI-D/ND: 102 ± 0 % Hamstrings PT: 208 ± 3 Nm PT/BW: 194 ± 1 Nm/kg LSI-D/ND: 101 ± 0 % Ratios HQR: 68 ± 1 %	Quadriceps PT/BW: 252 ± 28 Nm/kg Hamstrings PT/BW: 175 ± 27 Nm/kg Ratios HQR: 70 ± 9 %	Quadriceps PT: 136 ± 40 Nm PT/BW: 193 ± 7 Nm/kg LSI-D/ND: 103 %* TW: 1813 ± 480 (15 reps, range 90°) J LSI-D/ND (TW): 102 %* Hamstrings PT: 94 ± 29 Nm PT/BW: 129 ± 5 Nm/kg LSI-D/ND: 104 %* TW: 1596 ± 486 (15 reps, range 90°) J LSI-D/ND (TW): 99 %* Ratios HQR: 68 ± 1 %	Quadriceps MIVC/BW: 417 ± 56 N/kg Hamstrings MIVC/BW: 186 ± 24 N/kg Ratios HQR: 45 %*
	Non-elite	61 young adults, 23.7 ± 0.4 years old.[9,42]	Quadriceps PT/BW: 270 ± 18 Nm/kg LSI-D/ND: 105 ± 3 % Hamstrings PT/BW: 158 ± 4 Nm/kg LSI-D/ND: 102 ± 2 % Ratios HQR: 55 ± 10 %		Quadriceps PT/BW: 133 ± 25 Nm/kg LSI-D/ND: 106 %* Hamstrings PT/BW: 86 ± 23 Nm/kg LSI-D/ND: 106 %* Ratios HQR: 65 ± 15 %	
Female	Elite	34 young adults, 21.0 ± 0.0 years old.[14,28]	Quadriceps PT: 180 ± 42 Nm Hamstrings PT: 92 ± 18 Nm Ratios		Quadriceps PT: 87 ± 18 Nm Hamstrings PT: 62 ± 11 Nm Ratios	Quadriceps MIVC/BW: 396 ± 45 N/kg Hamstrings MIVC/BW: 166 ± 22 N/kg Ratios

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			HQR: 53 ± 10 %		HQR: 72 ± 10 %	HQR: 42 %*
	Non-elite	53 adolescents, 19.4 ± 1.3 years old.[12]	Ratios HQR: 63 ± 8 %		Ratios HQR: 74 ± 15 %	
		37 young adults, 22.6 ± 1.5 years old.[1,47]	Quadriceps PT: 123 ± 8 Nm PT/BW: 227 ± 27 Nm/kg Hamstrings PT: 49 ± 10 Nm PT/BW: 96 ± 22 Nm/kg	Quadriceps PT: 86 ± 5 Nm PT/BW: 160 ± 22 Nm/kg Hamstrings PT: 39 ± 8 Nm PT/BW: 73 ± 17 Nm/kg	Quadriceps PT: 61 ± 8 Nm PT/BW: 120 ± 15 Nm/kg Hamstrings PT: 30 ± 7 Nm PT/BW: 58 ± 14 Nm/kg	

HQR= concentric hamstring strength to concentric quadriceps strength ratio; LSI-D/ND=limb symmetry index calculated as ((value dominant leg)/(value non-dominant leg))*100%; MIVC/BW: maximum isometric voluntary contraction normalized to body weight; PT=peak torque; PT/BW=peak torque normalized to body weight; reps=repetitions; TW=total work.
*Standard deviation not available.

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TABLE 6. Quadriceps and hamstring strength reference values for pivoting sport athletes during rehabilitation after ACLR.[7,27,30,42,56,61]

Time point after ACLR	Sex	Sport level	N, age	Graft type	Isokinetic 60°/sec	Isokinetic 180°/sec	Isokinetic 300°/sec	Isometric 45°
3 months	Male	Non-elite	156 adolescents, 18.8 ± 3.1 years old.[27]	Q	Quadriceps PT/BW O: 141 ± 44 Nm/kg PT/BW NO: 246 ± 58 Nm/kg LSI: 58 ± 17 % Hamstrings PT/BW O: 118 ± 36 Nm/kg PT/BW NO: 140 ± 48 Nm/kg LSI: 86 ± 19 %			
	Female	Non-elite	164 adolescents, 17.4 ± 2.8 years old.[27]	Q	Quadriceps PT/BW O: 113 ± 54 Nm/kg PT/BW NO: 228 ± 42 Nm/kg LSI: 48 ± 16 % Hamstrings PT/BW O: 93 ± 27 Nm/kg PT/BW NO: 120 ± 31 Nm/kg LSI: 79 ± 22 %			
4 months	Male	Elite	20 young adults, 24.2 ± 5.1 years old.[30]	BPTB	Quadriceps PT O: 101 ± 35 Nm PT NO: 176 ± 38 Nm LSI: 57 %* Hamstrings PT O: 92 ± 24 Nm PT NO: 107 ± 21 Nm LSI: 86 %*	Quadriceps PT O: 84 ± 22 Nm PT NO: 126 ± 26 Nm LSI: 67 %* Hamstrings PT O: 76 ± 17 Nm PT NO: 85 ± 17 Nm LSI: 89 %*		Quadriceps MIVC O: 142 ± 48 N MIVC NO: 213 ± 55 N LSI: 67 %* Hamstrings MIVC O: 107 ± 21 N MIVC NO: 105 ± 26 N LSI: 102 %*
		Non-elite	38 young adults, 24.2 ± 4.7 years old.[56]	BPTB & H	Quadriceps PT O: 189 ± 52 Nm PT/BW O: 240 ± 50 Nm/kg PT NO: 262 ± 58 Nm PT/BW NO: 330 ± 50 Nm/kg LSI: 72 ± 12 % Hamstrings PT O: 128 ± 31 Nm PT NO: 143 ± 31 Nm LSI: 89 ± 14 %			

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6 months	Male	Elite	20 young adults, 24.2 ± 5.1 years old.[30]	BPTB	Quadriceps PT O: 122 ± 37 Nm PT NO: 179 ± 40 Nm LSI: 68 %* Hamstrings PT O: 99 ± 24 Nm PT NO: 111 ± 21 Nm LSI: 89 %*	Quadriceps PT O: 99 ± 31 Nm PT NO: 129 ± 32 Nm LSI: 77 %* Hamstrings PT O: 79 ± 16 Nm PT NO: 84 ± 14 Nm LSI: 94 %*		Quadriceps MIVC O: 165 ± 40 N MIVC NO: 225 ± 50 N LSI: 73 %* Hamstrings MIVC O: 111 ± 21 N MIVC NO: 110 ± 22 N LSI: 101 %*
		Non-elite	156 adolescents, 18.8 ± 3.1 years old.[27]	Q	Quadriceps PT/BW O: 203 ± 51 Nm/kg PT/BW NO: 279 ± 56 Nm/kg LSI: 72 ± 15 % Hamstrings PT/BW O: 145 ± 34 Nm/kg PT/BW NO: 152 ± 34 Nm/kg LSI: 95 ± 17 %			
			118 young adults, 23.6 ± 5.8 years old.[42]	BPTB	Quadriceps PT/BW O: 200 ± 45 Nm/kg PT/BW NO: 260 ± 45 Nm/kg LSI: 77 ± 14 % Hamstrings PT/BW O: 146 ± 29 Nm/kg PT/BW NO: 151 ± 28 Nm/kg LSI: 97 ± 12 %			
	Female	Non-elite	164 adolescent, 17.4 ± 2.8 years old.[27]	Q	Quadriceps PT/BW O: 161 ± 45 Nm/kg PT/BW NO: 253 ± 45 Nm/kg LSI: 63 ± 16 % Hamstrings PT/BW O: 122 ± 25 Nm/kg PT/BW NO: 135 ± 31 Nm/kg LSI: 91 ± 18 %			

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7months	Male	Non-elite	60 young adults, 25.8 ± 2.3 years old.[56,61]	BPTB & H for 60°/sec, BPTB for 180 and 300°/sec	Quadriceps PT O: 223 ± 51 Nm PT/BW O: 290 ± 50 Nm/kg PT NO: 267 ± 58 Nm PT/BW NO: 330 ± 50 Nm/kg LSI: 85 ± 13 % Hamstrings PT O: 144 ± 30 Nm PT NO: 149 ± 34 Nm LSI: 98 ± 8 %	Quadriceps LSI: 80 ± 12 % Hamstrings LSI: 102 ± 11 %	Quadriceps LSI: 82 ± 11 % Hamstrings LSI: 102 ± 27 %	
9 months	Male	Non-elite	298 young adults, 24.2 ± 4.6 years old.[7]	BPTB, H & Q	Quadriceps PT/BW O: 224 ± 47 Nm/kg LSI: 84 ± 14 % Hamstrings PT/BW O: 153 ± 29 Nm/kg LSI: 97 ± 14 %			
10 months	Male	Non-elite	38 young adults, 24.2 ± 4.7 years old.[56]	BPTB & H	Quadriceps PT O: 257 ± 51 Nm PT/BW O: 320 ± 60 Nm/kg PT NO: 270 ± 61 Nm PT/BW NO: 340 ± 50 Nm/kg LSI: 94 ± 15 % Hamstrings PTO: 150 ± 31 Nm PT NO: 153 ± 34 Nm LSI: 98 ± 8 %			

ACLR=anterior cruciate ligament reconstruction; BPTB=bone-patellar tendon-bone; H=hamstring; LSI=limb symmetry index calculated as ((value operated leg)/(value non-operated leg))*100%; MIVC=maximum isometric voluntary contraction; NO=non-operated leg; O=operated leg; PT=peak torque; PT/BW=peak torque normalized to body weight; Q=quadriceps.
*Standard deviation not available.

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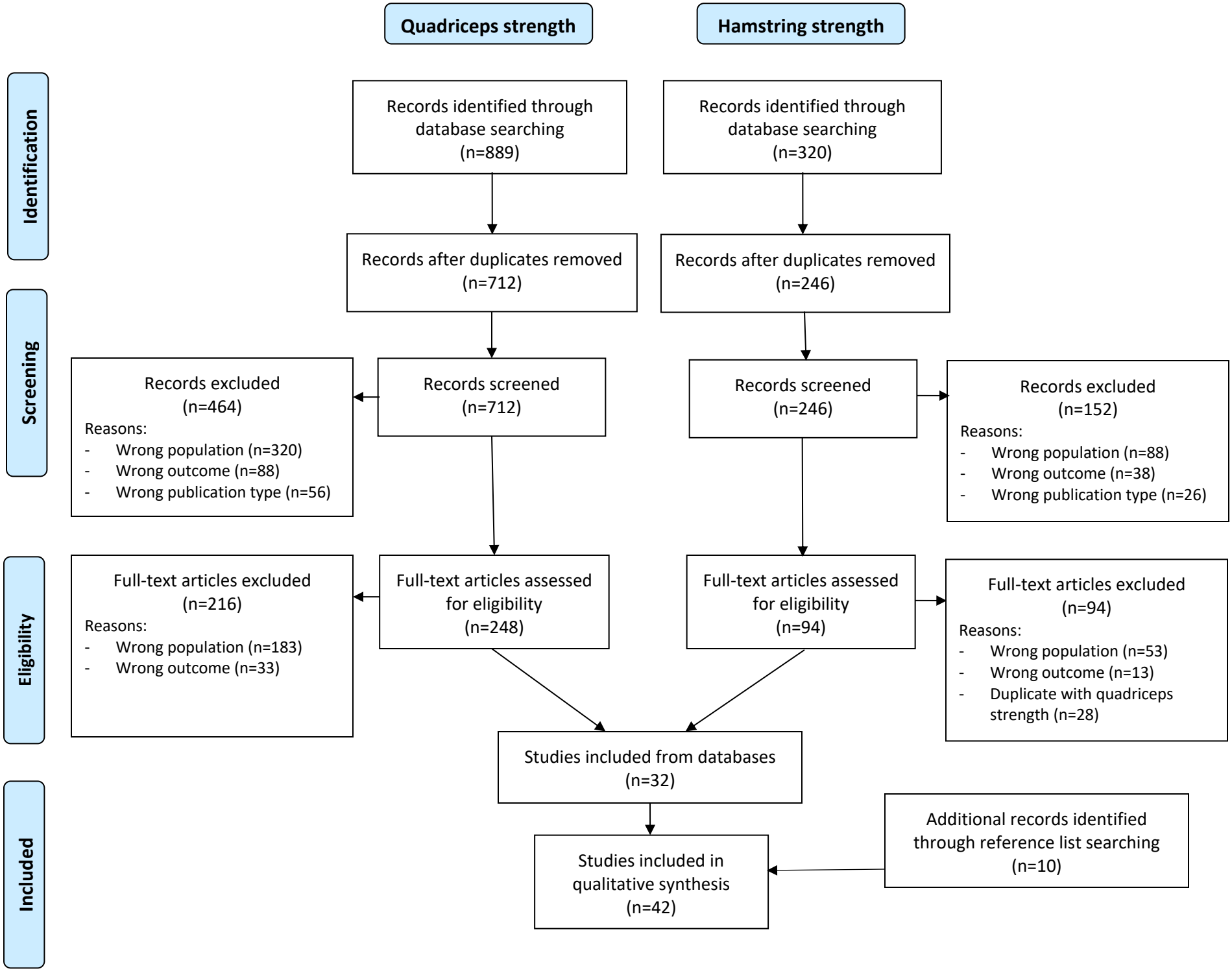

FIGURE 1. PRISMA flow chart of inclusion process.

Pubmed / MEDLINE / Cochrane Syntax

Quadriceps strength

(((((("quadriceps muscle"[Mesh] OR "quadriceps muscle"[TIAB]) OR "quadriceps"[TIAB]) OR "knee extensor"[TIAB]) OR (("movement"[Mesh] OR "movement"[All Fields]) AND “quantity”[All Fields])) AND ((((((((((("muscle strength"[Mesh] OR "muscle strength"[TIAB]) OR "strength test"[TIAB]) OR "torque"[Mesh]) OR "torque"[TIAB]) OR "torques"[TIAB]) OR "muscle power"[TIAB]) OR "muscle strength dynamometer"[Mesh]) OR "dynamometer"[TIAB]) OR "dynamometry"[TIAB]) OR "isokinetic"[TIAB]) OR "isometric"[TIAB])) AND ((((((((((((((("reference values"[Mesh] OR "reference values"[TIAB]) OR "reference value"[TIAB]) OR "normative value"[TIAB]) OR "normative values"[TIAB]) OR "normative data"[TIAB]) OR "symmetry index"[TIAB]) OR "limb difference"[TIAB]) OR "limb differences"[TIAB]) OR “absolute value”[TIAB]) OR "absolute values"[TIAB]) OR "normal range"[TIAB]) OR "normal ranges"[TIAB]) OR "normal value"[TIAB]) OR "normal values"[TIAB]) OR "reference range"[TIAB]) OR "reference ranges"[TIAB])

Hamstring strength

(((((("hamstring muscles"[Mesh] OR “hamstring muscles”[TIAB]) OR "hamstring"[TIAB]) OR "knee flexor"[TIAB]) OR (("movement"[Mesh] OR "movement"[All Fields]) AND “quantity”[All Fields])) AND ((((((((((("muscle strength"[Mesh] OR "muscle strength"[TIAB]) OR "strength test"[TIAB]) OR "torque"[Mesh]) OR "torque"[TIAB]) OR "torques"[TIAB]) OR "muscle power"[TIAB]) OR "muscle strength dynamometer"[Mesh]) OR "dynamometer"[TIAB]) OR "dynamometry"[TIAB]) OR "isokinetic"[TIAB]) OR "isometric"[TIAB])) AND ((((((((((((((("reference values"[Mesh] OR "reference values"[TIAB]) OR "reference value"[TIAB]) OR "normative value"[TIAB]) OR "normative values"[TIAB]) OR "normative data"[TIAB]) OR "symmetry index"[TIAB]) OR "limb difference"[TIAB]) OR "limb differences"[TIAB]) OR “absolute value”[TIAB]) OR "absolute values"[TIAB]) OR "normal range"[TIAB]) OR "normal ranges"[TIAB]) OR "normal value"[TIAB]) OR "normal values"[TIAB]) OR "reference range"[TIAB]) OR "reference ranges"[TIAB])

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Embase syntax

Quadriceps strength

((("quadriceps muscle" OR "quadriceps") OR "knee extensor") OR ("movement" AND “quantity”)) AND (((((((("muscle strength" OR "strength test") OR "torque") OR "torques") OR "muscle power") OR "muscle strength dynamometer") OR "dynamometer") OR "dynamometry") OR "isokinetic") OR "isometric") AND ((((((((((((("reference values" OR "reference value") OR "normative value") OR "normative values") OR "normative data") OR "symmetry index") OR "limb difference") OR "limb differences") OR “absolute value”) OR "absolute values") OR "normal range") OR "normal ranges") OR "normal value") OR "normal values") OR "reference range") OR "reference ranges")

Hamstring strength

((("hamstring muscles" OR "hamstring") OR "knee flexor") OR (“movement" AND “quantity”[ALL FIELDS])) AND (((((((("muscle strength" OR "strength test") OR "torque") OR "torques") OR "muscle power") OR "muscle strength dynamometer") OR "dynamometer") OR "dynamometry") OR "isokinetic") OR "isometric") AND ((((((((((((("reference values" OR "reference value") OR "normative value") OR "normative values") OR "normative data") OR "symmetry index") OR "limb difference") OR "limb differences") OR “absolute value”) OR "absolute values") OR "normal range") OR "normal ranges") OR "normal value") OR "normal values") OR "reference range") OR "reference ranges")

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Web of Science syntax

Quadriceps strength

(quadriceps OR “knee extensor” OR “movement quantity”) AND (“muscle strength” OR “strength test” OR torque OR torques OR “muscle power” OR dynamometer OR dynamometry OR isokinetic OR isometric) AND (“reference value” OR “reference values” OR “normative value” OR “normative values” OR “normative data” OR “symmetry index” OR “limb difference” OR “limb differences” OR “absolute value” OR “absolute values” OR “normal range” OR “normal ranges” OR “normal value” OR “normal values” OR “reference range” OR “reference ranges”)

Hamstring strength

(hamstring OR “knee flexor” OR “movement quantity”)
AND
(“muscle strength” OR “strength test” OR torque OR torques OR “muscle power” OR dynamometer OR dynamometry OR isokinetic OR isometric)
AND
(“reference values” OR “reference value” OR “normative value” OR “normative values” OR “normative data” OR “symmetry index” OR “limb difference” OR “limb differences” OR “absolute value” OR “absolute values” OR “normal range” OR “normal ranges” OR “normal value” OR “normal values” OR “reference range” OR “reference ranges”)

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Study characteristics of all 42 included articles about quadriceps and hamstring strength

Author, year and country of publication	Population	Test	Test conditions	Test outcomes
Agopyan et al. 2018, Turkey[1]	Healthy female modern dancers (n=20), non-elite level; 24±4 years old, 51±4 kg, 162±7 cm.	concentric Q + H at 60 (4 reps), 180 (4 reps) & 300°/sec (30 reps); range not specified; Biodex 3.	Between 2 and 5 p.m. with no exercise 48 hours before	PT PT/BW
Bittencourt et al. 2005, Brazil[3]	Healthy male volleyball players (n=20), elite level U19 team; 17±1 years old, 86±9 kg, 197±5 cm; Healthy male volleyball players (n=16), elite level U21 team; 20±1 years old, 92±7 kg, 197±6 cm.	concentric Q + H at 60 (5 reps) & 300°/sec (30 reps); range 110-0° of flexion; Biodex 3.	Not specified	PT/BW LSI-D/ND HQR
Bogdanis et al. 2015, Greece[5]	Healthy male soccer players (n=18), elite level; 24±1 years old, 77±1 kg, 182±1 cm.	concentric Q + H at 60, 180 & 300°/sec (all 3 reps); range 100-0° of flexion; Biodex.	Mid-season	PT PT/BW HQR
Carolan et al. 2020, Ireland[7]	Male pivoting athletes with ACLR with H, BPTB or Q graft (n=298), non-elite level; 24±5 years old, 83±10 kg, 180±6 cm.	concentric Q + H at 60°/sec (5 reps); range 100-0° of flexion; Humac Norm.	9 months post-ACLR	PT/BW LSI
Carvalho et al. 2016, Portugal[8]	Healthy male soccer players (n=159), elite level; 26±5 years old, 75±7 kg, 180±1 cm.	concentric Q + H at 60°/sec, eccentric Q + H at 60°/sec (all 5 reps); range 90-20° of flexion; Technogym REV 9000.	Pre-season, between 9:30 and 11:30 a.m. with no exercise 24 hours before	PT HQR DCR
Cheung et al. 2012, China[9]	Healthy male soccer players (n=23), non-elite level; 22±2 years old, 65±5 kg, 174±5 cm. Healthy male basketball and volleyball players (n=17), non-elite level; 23±1 years old, 69±8 kg, 175±7 cm.	concentric Q + H at 60 & 300°/sec (both 5 reps); range not specified; Cybex Norm.	During the season, no vigorous exercise 48 hours before	PT/BW LSI-D/ND HQR
Cometti et al. 2001, France[10]	Healthy male soccer players (n=63), elite level; 25±5 years old, 75±15 kg, 179±6 cm. Healthy male soccer player (n=32), non-elite level; 26±4 years old, 77±18 kg, 178±5 cm.	concentric Q + H at 60, 180 & 300°/sec, eccentric Q + H at 60°/sec (all 3 reps); range 90-0° of flexion; Biodex.	End-season	PT HQR DCR

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Daneshjoo et al. 2012, Malaysia[11]	Healthy male soccer players (n=36), elite level; 20±2 years old, 76±6 kg, 183±5 cm.	concentric Q + H at 60, 180 & 300°/sec, number of reps not specified; range 90-0° of flexion; Biodex 3.	Mid-season, between 8 and 11 a.m.	HQR
Devan et al. 2004, USA[12]	Healthy female hockey, soccer and basketball players (n=53), non-elite level; 19±1 years old, 65±10 kg, 168±10 cm.	concentric Q + H at 60 (5 reps) & 300°/sec (15 reps); range not specified; Biodex.	Pre-season	HQR
van Dyk et al. 2017, Qatar[13]	Healthy male soccer players (n=347), elite level; 26±5 years old, 73±9 kg, 177±7 cm.	concentric Q + H at 60 (5 reps) & 300°/sec (10 reps), eccentric H at 60°/sec (5 reps); range not specified; Biodex 3.	End-season	PT PT/BW
Dos Santos Andrade et al. 2012, Brazil[14]	Healthy male jodukas (n=32), elite level; 24±4 years old, 84±24 kg, 177±10 cm. Healthy female judokas (n=26), elite level; 21±3 years old, 68±17 kg, 161±21 cm. Healthy male handball players (n=17), elite level; 26±4 years old, 94±12 kg, 187±8 cm. Healthy female handball players (n=22), elite level; 25±3 years old, 74±8 kg, 177±7 cm. Healthy male soccer players (n=52), elite level; 25±4 years old, 75±8 kg, 178±7 cm. Healthy female soccer players (n=17), elite level; 25±7 years old, 60±7 kg, 163±5 cm.	concentric Q + H at 60 & 300°/sec (both 5 reps); range 95-5° of flexion; Biodex.	Pre-season	PT HQR
Eniseler et al. 2012, Turkey[15]	Healthy male soccer players (n=14), elite level; 26±4 years old, 75±3 kg, height not specified.	concentric Q + H at 60 & 300°/sec (both 4 reps); range 90-0° of flexion; Biodex 3.	Pre-season	PT LSI-D/ND HQR
Forbes et al. 2009, UK[16]	Healthy male soccer players (n=29), elite level; 17±1 years old, 70±6 kg, 179±5 cm.	concentric Q + H at 60°/sec (5 reps), eccentric H at 60°/sec (5 reps); range not specified; Biodex 3.	Not specified	PT LSI-D/ND DCR

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Grammatikopoulou et al. 2005, Greece[17]	Healthy female basketball (n=14), elite level; 24±3 years old, 67±7 kg, 180±6 cm; Healthy female handball (n=14), elite level; 24±3 years old, 57±7 kg, 165±3 cm.	concentric Q + H at 60, 180 & 300°/sec (all 3 reps); range not specified; Cybex II.	End-season	PT PT/BW HQR
Greig et al. 2008, UK[18]	Healthy male soccer players (n=10), elite level; 25±4 years old, 77±8 kg. height not specified.	concentric Q + H at 60, 180 & 300°/sec (all 5 reps), eccentric H at 60, 180 & 300°/sec (all 5 reps); range 90-0° of flexion; Biodex 3.	Between 2 and 5 p.m.	PT HQR DCR
Grygorowicz et al. 2017, Poland[21]	Healthy male soccer players (n=55), elite level; 23±5 years old, 78±7 kg, 183±5 cm.	concentric Q + H at 60°/sec (3 reps); range 90-0° of flexion; Biodex 3.	Pre-season	PT PT/BW HQR
Gür et al. 1999, Turkey[22]	Healthy male soccer players (n=13), elite level; 25±3 years old, 72±4 kg, 177±5 cm. Healthy male soccer players (n=12), elite level; 19±1 years old, 64±5 kg, 175±6 cm.	concentric Q + H at 180 & 300°/sec, eccentric Q + H at 180 & 300°/sec (all 4 reps); range 90-10° of flexion; Cybex 6000.	Not specified	PT LSI-D/ND HQR DCR
Hoshikawa et al. 2009, Japan[26]	Healthy male soccer players (n=20), elite level; 23±2 years old, 70±6 kg, 176±5 cm.	concentric Q + H at 60 (3 reps) & 180°/sec (5 reps); range not specified; Biodex 3.	Pre-season	PT LSI-D/ND HQR
Hunnicutt et al. 2020, USA[27]	Male pivoting athletes with ACLR with Q (n=156), non-elite level; 19±3 years old, 81±14 kg, 179±9 cm. Female pivoting athletes with ACLR with Q (n=164), non-elite level; 17±3 years old, 63±10 kg, 167±8 cm.	concentric Q + H at 60°/sec (5 reps); range not specified; Biodex 4.	3 and 6 months post-ACLR	PT/BW LSI
Jordan et al. 2015, Canada[28]	Healthy male skiers (n=13), elite level; 22±3 years old, 84±7 kg, height not specified; Healthy female skiers (n=8), elite level; 21±2 years old, 65±2 kg, height not specified.	isometric Q + H, duration 2s; 70° of flexion; Biodex 3.	Pre-season	MIVC/BW
Knezevic et al. 2014, Serbia[30]	Male soccer players, handball players and judokas before and with ACLR with BPTB graft (n=20), elite level;	concentric Q + H at 60 & 180°/sec (both 5 reps); range 90-10° of flexion;	4 and 6 months post-ACLR	PT LSI MIVC

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	24±5 years old; 84±11kg, 180±4 cm.	isometric Q + H, duration 3-4s; 45° of flexion; Kin-Com AP125.		LSI (MIVC)
Kobayashi et al. 2006, Japan[31]	Healthy female basketball players (n=12), non-elite level; 20±0 years old, 56±2 kg, 167±2 cm; Healthy male soccer players (n=10), non-elite level; 18±0 years old, 62±2 kg, 171±2 cm.	concentric H + Q at 180 & 300°/sec (both 5 reps); range not specified; Cybex.	In the morning	PT PT/BW HQR
Lund-Hanssen et al. 1996, Norway[33]	Healthy female handball players (n=114), elite level; 20 (16-27) years old, 64 (48-82) kg, 170 (155-183) cm.	concentric H + Q at 60°/sec (5 reps); range 90-0° of flexion; Biodex.	Not specified	PT HQR
Metaxas et al. 2009, Greece[38]	Healthy male soccer players (n=100), elite level; 25±5 years old, 75±7 kg, 178±6 cm; Healthy male basketball players (n=61), elite level; 23±4 years old, 93±13 kg, 191±8 cm.	concentric Q + H at 60, 180 & 300°/sec (all 3 reps); range not specified; Cybex Norm.	Not specified	PT HQR
Minozzo et al. 2018, Brazil[39]	Healthy male soccer players (n=93), elite level; 26±5 years old, 78±8 kg, 181±7 cm.	concentric Q + H at 60°/sec, eccentric H at 60°/sec (all 5 reps); range 90-0° of flexion; Cybex Norm.	Pre-season	PT HQR DCR
Nunes et al. 2016, Brazil[41]	Healthy male futsal players (n=11), elite level; 24±2 years old, 79±10 kg, 178±3 cm.	concentric Q + H at 60°/sec (3 reps); range 90-0° of flexion; Cybex Norm.	Initial phase of competitive season with no heavy exercise 48 hours before	PT PT/BW
O'Malley et al. 2018, Ireland[42]	Male pivoting athletes with ACLR with BPTB graft (n=118), non-elite level; 24±6 years old, 82±11 kg, 183±7 cm. Healthy male pivoting athletes (n=44), non-elite level; 24±4 years old, 83±8 kg, 183±7 cm.	concentric Q + H at 60°/sec (5 reps); range not specified; Cybex Norm.	6 months post-ACLR; Not specified for healthy athletes	PT/BW LSI LSI-D/ND
Orchard et al. 1997, Australia[43]	Healthy male Australian rules football players (n=37), elite level; 22±4 years old, 86±9 kg, 185±7 cm.	concentric Q + H at 60, 180 & 300°/sec (all 3 reps); range not specified; Cybex 340.	Pre-season	PT/BW HQR
Ostenberg et al. 1998, Sweden[44]	Healthy female soccer players (n=101), non-elite level; 20±4 years old, 61±7 kg, 167±5 cm.	concentric Q at 60 & 180°/sec (both 4 reps); range 100-0° of flexion; Cybex II.	Pre-season	PT LSI-D/ND

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Padulo et al. 2020, Italy[46]	Healthy male soccer players (n=17), elite level; 16±1 years old, 72±9 kg, 175±6 cm.	isometric Q + H, duration 4s; 90° of flexion; Biodex 3.	Between 3 and 5 p.m.	MIVC
Pua et al. 2007, Singapore[47]	Healthy female pivoting athletes (n=17), non-elite level; 21±2 years old, 51±5 kg, 161±5 cm.	concentric Q at 60 & 180°/sec (both 6 reps); range 90-10° of flexion; Biodex 3.	Not specified	PT
Risberg et al. 2018, Norway[49]	Healthy female soccer players (n=196), elite level; 21±4 years old, 64±8 kg, 167±6 cm; Healthy female handball players (n=143), elite level; 21±4 years old, 69±8 kg, 172±6 cm.	concentric Q + H at 60°/sec (3 reps); range 90-15° of flexion; Technogym REV 9000.	Pre-season	PT PT/BW LSI-D/ND HQR
Ruas et al. 2015, Brazil[50]	Healthy male soccer players (n=102), elite level; 26±5 years old, 79±8 kg, 181±7 cm.	concentric Q + H at 60°/sec, eccentric H at 60°/sec (all 5 reps); range 90-0° of flexion; Cybex Norm.	Pre-season	PT LSI-D/ND HQR DCR
Śliwowski et al. 2017, Poland[51]	Healthy male soccer players (n=111), elite level; 26±5 years old, 79±8 kg, 181±7 cm.	concentric Q + H at 60 (3 reps); range 90-0° of flexion; Biodex 3.	Between 9 a.m. and 1 p.m. with no intensive training 48 hours before	PT/BW LSI-D/ND HQR
Theoharopoulos et al. 2000, Greece[52]	Healthy male basketball players (n=12), elite level; 21±3 years old, 88±15 kg, 191±11 cm.	concentric Q at 60 (6 reps) & 180°/sec (9 reps); range 100-0° of flexion; Cybex 1200.	End-season	PT PT/BW LSI-D/ND HQR
Tourny-Chollet et al. 2002, France[53]	Healthy male soccer players (n=21), non-elite level; 22±3 years old, 75±7 kg, 180±7 cm.	concentric Q + H at 60°/sec, eccentric Q + H at 60°/sec (all 3 reps); range 80-0° of flexion; Kin-Com 500H.	Not specified	PT/BW HQR DCR
Voutselas et al. 2007, Greece[55]	Healthy male soccer players (n=72), elite level; 20±5 years old, 70±12 kg, 173±22 cm.	concentric Q + H at 60°/sec (3 reps); range 90-0° of flexion; Cybex Norm.	Not specified	PT PT/BW LSI-D/ND HQR
Welling et al. 2019, the Netherlands[56]	Male soccer players with ACLR with BPTB or H graft (n=38), non-elite level; 24±5 years old, 79±13 kg, height not specified. Healthy male soccer players (n=30), non-elite level; 23±3 years old, 73±7 kg, height not specified.	concentric Q + H at 60°/sec (5 reps); range 100-0° of flexion; Biodex 3.	4, 7 and 10 months post-ACLR; Not specified for healthy athletes	PT PT/BW LSI LSI-D/ND

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Wik et al. 2018, Qatar[59]	Healthy male soccer players (n=195), elite level; 24±5 years old, 72±9 kg, 176±7 cm.	concentric Q + H at 60°/sec, eccentric H at 60°/sec (all 5 reps); range not specified; Biodex.	Not specified	PT PT/BW LSI-D/ND DCR
Xergia et al. 2015, Greece[61]	Male pivoting athletes with ACLR with BPTB graft (n=22), non-elite level; 29±11 years old, 77±11 kg, 177±4 cm.	concentric Q + H at 180 & 300°/sec (both 5 reps); range 90-0° of flexion; Biodex 3.	7 months post-ACLR	LSI
Zabka et al. 2011, Brazil[62]	Healthy male soccer players (n=39), elite level; 24±4 years old, 78±8 kg, 182±7 cm.	concentric Q + H at 60°/sec (4 reps); range 100-0° of flexion; Biodex 4.	Not specified	PT/BW HQR
Zvijac et al. 2014, USA[63]	Healthy male American football players (n=1252), elite level; age 20-27 years old, 110±21 kg, height not specified.	concentric Q + H at 60 (3 reps) & 300°/sec (15 reps); range 90-0° of flexion; Cybex Norm.	Not specified	PT PT/BW LSI-D/ND TW HQR

ACL=anterior cruciate ligament; ACLR=ACL reconstruction; BPTB=bone-patellar tendon-bone; DCR=dynamic control ratio=eccentric hamstring strength to concentric quadriceps strength ratio; H=hamstrings; HQR=concentric hamstring strength to concentric quadriceps strength ratio; LSI=limb symmetry index; LSI-D/ND=limb symmetry index dominant/non-dominant; MIVC=maximum isometric voluntary contraction; MIVC/BW=maximum isometric voluntary contraction normalized to body weight; PT=peak torque; PT/BW=peak torque normalized to body weight; Q=quadriceps; TW=total work; reps=repetitions.